

Module 04: IPv4 Classful Addressing & Special Purpose Blocks

Complete mathematical exploration of IPv4 Class A, B, C, D, and E architectures, NetID vs HostID divisions, default subnet masks, usable host calculations ($2^H - 2$), and RFC 1918 private address ranges.

IPv4 Classful Addressing Hierarchy: Classes A, B, C, D & E Breakdown

1. Class Bit Patterns & Address Ranges

0	NetID (7b)	HostID (24 bits = 16,777,214 Hosts)
10	NetID (14 bits = 16,384 Nets)	HostID (16 bits = 65,534 Hosts)
110	NetID (21 bits = 2,097,152 Nets)	Host (8b: 254)
1110	Multicast Group Address (28 bits)	
1111	Reserved for Future / Research & Development	

Default Subnet Masks:
Class A: 255.0.0.0 (/8) | Class B: 255.255.0.0 (/16) | Class C: 255.255.255.0 (/24)

2. Class Decimal Ranges & Special Blocks

Class A	0.0.0.0 - 127.255.255.255	Leading bits: 0...
Class B	128.0.0.0 - 191.255.255.255	Leading bits: 10...
Class C	192.0.0.0 - 223.255.255.255	Leading bits: 110...
Class D	224.0.0.0 - 239.255.255.255	Multicast Groups
Class E	240.0.0.0 - 255.255.255.255	Experimental / R&D

RFC 1918 Private IPv4 Ranges (Non-Routable):

- Class A: 10.0.0.0 to 10.255.255.255 (10.0.0.0/8)
- Class B: 172.16.0.0 to 172.31.255.255 (172.16.0.0/12)
- Class C: 192.168.0.0 to 192.168.255.255 (192.168.0.0/16)

Conserves global IPv4 space; requires NAT for internet access!

Special Purpose IPv4 Addresses (Never Assigned to Regular Host NICs)

This Host (0.0.0.0):

Used by host during bootstrapping (DHCP Discover) when it doesn't yet know its own IP address.

Loopback (127.0.0.1 - 127.x.x.x):

Packets never leave the local machine. Used to test local TCP/IP stack software (localhost).

Limited Broadcast (255.255.255.255):

Broadcast to all hosts on the local network segment. Strictly NEVER forwarded by routers!

Directed Broadcast (HostID = All 1s):

E.g., 192.168.1.255. Delivered to all hosts in that specific remote network.

Network ID (HostID = All 0s):

E.g., 192.168.1.0. Identifies the entire network itself; cannot be assigned to an individual host interface.

1. The 5 Address Classes

Classful addressing divides the 32-bit address space based on leading bits: Class A (0, /8), Class B (10, /16), Class C (110, /24), Class D (1110, Multicast), and Class E (1111, Experimental).

2. Usable Host Capacity Equation

Address Exclusion Rules

Usable Hosts = $2^H - 2$

Excluded: Network Address (HostID = All 0s) and Directed Broadcast (HostID = All 1s)

Private IP Blocks: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16